

DATA DICTIONARY

Research Title	A multimodal dataset of touchscreen kinetics and inertial sensor logs during mindful and mindless smartphone scrolling
Principal Investigators	Leonard Wangka; Metta Kireina Tedjasurya
Affiliation	School of Computer Science, Bina Nusantara University

1. File Overview

The dataset is organized by participant IDs. Each participant folder contains four files representing two different cognitive states (Mindful and Mindless) and two types of logs (Sensor and Touch).

- File Naming Convention: [LogType]_P-[ID]_[Condition].csv

2. SensorLog Schema

These files contain high-frequency data from the device's movement sensors.

Column Name	Data Type	Description	Units/Values
Timestamp	Long	The Unix timestamp when the sensor event was recorded.	<i>ms</i>
Sensor_Type	String	The type of inertial sensor capturing the data.	<i>ACCEL, GYRO</i>
X	Double	Acceleration or angular velocity along the X-axis.	m/s^2 (<i>ACCEL</i>) rad/s (<i>GYRO</i>)
Y	Double	Acceleration or angular velocity along the Y-axis.	m/s^2 (<i>ACCEL</i>) rad/s (<i>GYRO</i>)
Z	Double	Acceleration or angular velocity along the Z-axis.	m/s^2 (<i>ACCEL</i>) rad/s (<i>GYRO</i>)

3. TouchLog Schema

These files capture the interaction events occurring on the device screen.

Column Name	Data Type	Description	Units/Values
Timestamp	Long	The Unix timestamp when the sensor event was recorded.	<i>ms</i>
Event_Type	String	The classification of the user's interaction.	<i>CLICK, SCROLL</i>
ScrollDeltaX	Double	The relative horizontal distance of the scroll/swipe movement.	Pixels
ScrollDeltaY	Double	The relative vertical distance of the scroll/swipe movement.	Pixels
PackageName	String	The Android package name of the active application.	<code>com.ss.android.ugc.trill</code> (TikTok)

4. Participant_Metadata Schema

This file provides demographic and physical context for each participant.

Column Name	Data Type	Description	Values
Participant_ID	String	Unique identifier for participant.	P-001 to P-020
Age	Integer	The age of the participant at the time of data collection.	Years
Gender	String	Gender of the participant.	Male (M), Female (F)
Handedness	String	The dominant hand used by the participant for smartphone interaction.	Right, Left